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Gravity fingering control on evaporation and deep drainage in a 3D porous medium

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ABSTRACT

Water infiltration in soils is a fundamental hydrologic process that determines the water availability for plant roots and the rate of recharge to deep groundwater bodies. When both the initial water saturation of the soil and the water infiltration rate is low, the infiltration front becomes unstable, resulting in gravity fingers that act as preferential flow channels. Preferential flow may allow for the rapid bypass of water through the evaporation zone, thereby promoting deep drainage of water into the subsoil even in arid and semi-arid regions, where the potential evapotranspiration (PET) is high and the mean annual precipitation (MAP) is low. Here, we investigate experimentally the impact of gravity fingering on deep drainage coupled with surface evaporation in a 3D porous medium. We use a cylindrical vessel packed with dry glass beads, and reproduce seasonal climatic patterns consisting of relatively short rainfall events alternating with periods of intense evaporation. We vary the volume of rainfall events and the interval between rainfall events to replicate climate conditions. We record the mass of water in the system throughout the experiment, and determine the partition between evaporation and deep drainage before and after the onset of fingering. We find that in semi-arid conditions, the fraction of water that bypasses the evaporation zone increases fourfold after gravity fingering starts. This finding has potentially important implications for understanding infiltration dynamics in arid environments and ecosystems' response in regions susceptible to desertification.

1. Introduction

Understanding the subsurface percolation of rainwater is of critical importance to the long-term availability of water resources. The interplay between climate, soil moisture and vegetation determines the sustainability of the water and nutrient cycles on which biodiversity and human development depend (Schlesinger et al., 1990; Brown et al., 1997; Scanlon et al., 2007; Kéfi et al., 2007; Jackson et al., 2002; Reynolds et al., 2007; Richter and Mobley, 2009; Báez and Collins, 2008; Tietjen, 2016). In addition, studies suggest that global warming may trigger desertification (Jackson et al., 2005; Rietkerk et al., 2004) and ecosystem shifts, stressing the importance of understanding the underlying mechanisms determining the availability of the water in the Earth's critical zone (Rodriguez-Iturbe, 2000).

In the hydrologic cycle, rainfall will either flow over the land as surface runoff, return to the atmosphere through evapotranspiration, or infiltrate deep into the ground, reaching deep root systems and recharging aquifers (Horton, 1933; Hillel, 1980; Allison et al., 1994; Sophocleous, 2002). We chose the term "deep drainage" as it seems to be the most commonly accepted term in hydrology (Seyfried et al., 2005; Eilers et al., 2007; Wohling et al., 2012; Zhang et al., 2014; Li et al., 2019) to denote the flux of water that moves through the unsaturated zone past the root zone and may become recharge after a lag time. To determine the partitioning of water, the convention is to conduct a soil–water balance (Rodriguez-Iturbe et al., 2001; Rodriguez-Iturbe and Porporato, 2004; Small, 2005). A critical but poorly understood question is how partitioning depends on the hydrodynamics of infiltrated water and soil infiltration patterns.

Here we investigate the impact of gravity fingering—a widely observed instability that takes place during infiltration in relatively dry soil (Hill and Parlange, 1972; Glass et al., 1989; Ritsema et al., 1998). This instability has been documented in the field (Ritsema and Dekker, 1994; Hendrickx and Flury, 2001), and has been extensively characterized in quasi-2D laboratory experiments (Diment and Watson, 1985; Glass et al., 1989; Selker et al., 1992; Selker et al., 1992; Lu et al., 1994; Bauters et al., 1998; Bauters et al., 2000; Yao and Hendrickx, 2001; Sililo

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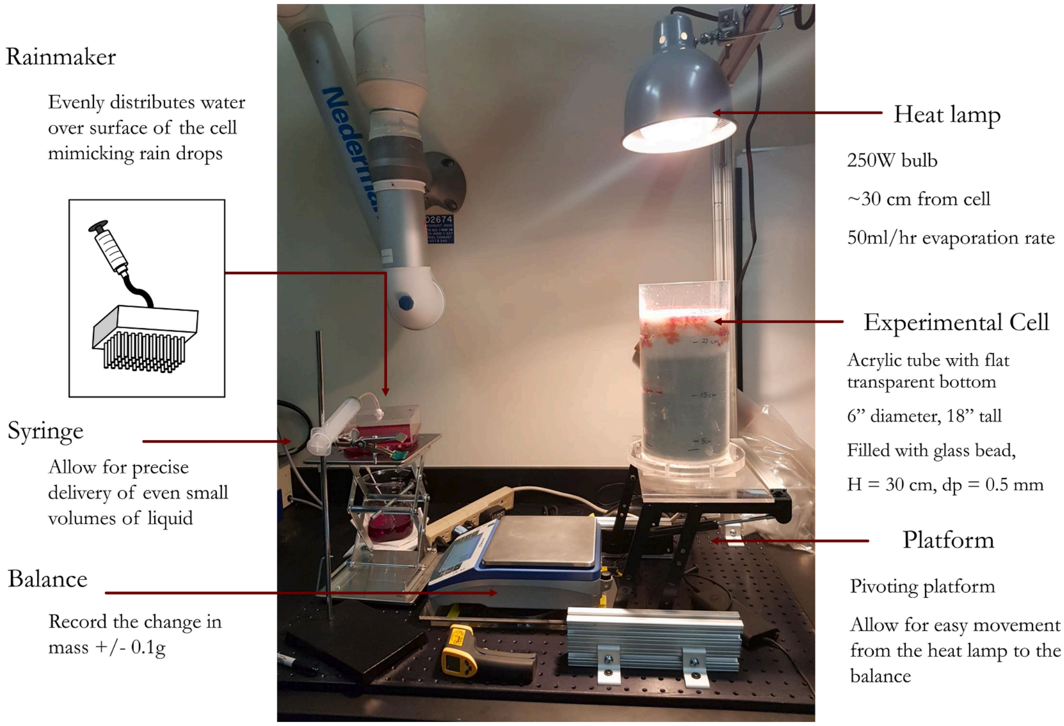

Fig. 1. Photo of the experimental setup showing the cell underneath a heat lamp. To the left of the cell is a balance. The cell sits on a movable platform so it can pivot back and forth between the heat lamp and the balance. Also depicted is a removable rain maker with attached syringe.

and Tellam, 2000; Flekkøy et al., 2002; Wang et al., 2004; Wei et al., 2014) and some 3D experiments (Yao and Hendrickx, 1996; Glass et al., 1990). Some of the key features of fingered flow are:

1. Presence of saturation overshoot, that is, the pile-up of water at the infiltration front, which is understood to be a prerequisite for the fingering instability in unsaturated flows (DiCarlo, 2004; Egorov et al., 2003; Nieber et al., 2005; DiCarlo, 2007; Cueto-Felgueroso and Juanes, 2009).
2. Winner-takes-all behavior, in which the fastest-growing fingers channelize most of the flow, and the growth of other incipient fingers is thereby suppressed (Glass et al., 1989; Selker et al., 1992).
3. Robustness of the instability with respect to the heterogeneity of the medium (Cremer et al., 2019; Sililo and Tellam, 2000; Cueto-Felgueroso et al., 2020).
4. Strong dependence on the infiltration flux, indicating that stable fronts are observed in dry media when the infiltration rate is either very small or approaches the saturated conductivity (Hendrickx and Yao, 1996). In general, larger infiltration rates produce faster, thicker fingers (Glass et al., 1989).
5. Fingering is promoted when the medium is quite dry; even relatively low water saturations lead to a compact, downward-moving wetting front (Lu et al., 1994; Bauters et al., 2000).

Experimental work has shown that the fingering instability is expected to be more vigorous for coarse soils, for two reasons: importance of gravity vs. capillary forces, and because the soil retention curve and unsaturated conductivity are often 'sharper' (more nonlinear) for sands than for clays.

It is unquestionable that soil heterogeneity, structure and texture have a strong impact on water infiltration. However, the presence of heterogeneity due to soil texture (including hydrophobicity) can in fact exacerbate the effects of the fingering hydrodynamic instability. Even the presence of soil layering—which could be intuitively understood as the most unfavorable configuration for the development of fingering—does not prevent the formation and persistence of gravity fingers.

Of course, gravity fingering is an instance of the more general phenomenon of preferential flow (Allaire et al., 2009). Among the various potential mechanisms for preferential flow, much attention has been devoted to the role of 'macropores', that is, large pores formed by soil fauna, plant roots, cracks and fissures, and erosion pipes (Beven and Germann, 1982). While 'macropores' can be significant drivers of soil water, most of the previous work has focused, almost exclusively, on clayey soil types and humid conditions. These are conditions of much less relevance for our proposed study, which focuses on the interaction between infiltration and evapotranspiration in arid and semi-arid environments. In fact, direct 3D imaging indicates the prevalence of 'macropore' geometry in preferential flow for clayey soil, and fingering for sandy soil (Heijs et al., 1996).

Several modeling approaches have been proposed to capture the dynamics of infiltration under gravity fingering. These include a phase-field continuum formulation that reproduces the experimental observations above in 2D (Cueto-Felgueroso and Juanes, 2008; Cueto-Felgueroso and Juanes, 2009; Cueto-Felgueroso et al., 2020) and 3D (Gomez et al., 2013), is endowed with an entropy function (Beljadid et al., 2020), and is rooted on the thermodynamics of two-phase flow at the pore scale (Cueto-Felgueroso and Juanes, 2012).

Despite this extensive body of knowledge on the dynamics of fingered flow in the unsaturated zone, its impact on evaporation and deep drainage remains unexplored. We hypothesize that gravity fingering can lead to bypassing the evaporation in the first few centimeters of soil. The transition from compact infiltration to fingered flow could then exert a fundamental control on the partition between evapotranspiration and deep drainage, thereby modulating long-term recharge in arid and semi-arid regions, ecosystems' hydrologic functions, and their resilience to water stress and climatic change. Here we investigate this question by means of laboratory experiments that couple water infiltration in a fully 3D porous medium with water evaporation under experimental conditions that replicate environmentally-relevant climatic, rainfall and soil conditions in the field.

The paper is organized as follows. In Section 2, we present a novel experimental setup which enables the study of infiltration coupled to

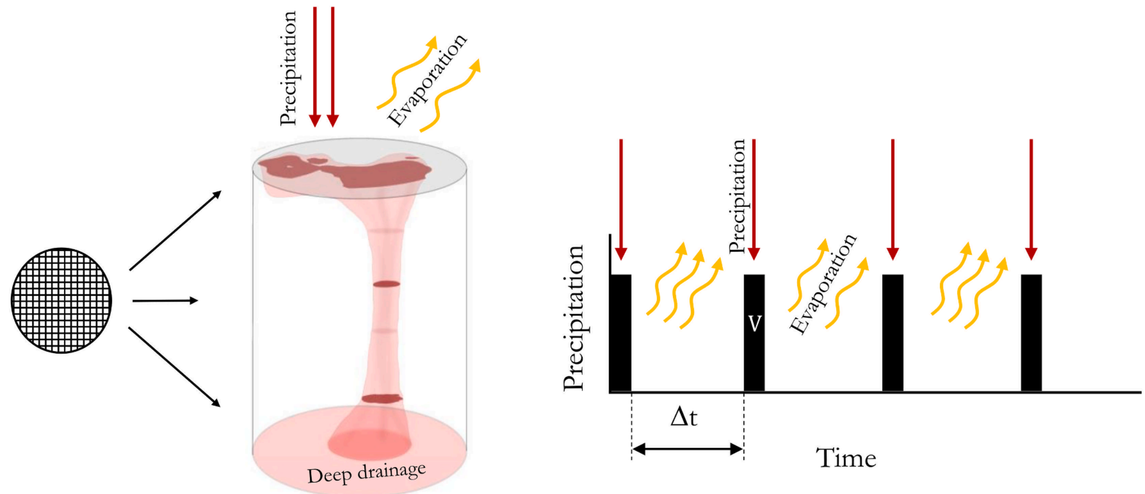

Fig. 2. Left: Schematic of the 3D experimental cell, depicting precipitation and evaporation at the surface of the cell, and illustrating a reconstruction of the finger morphology from the locations of the nylon mesh. The darker read areas of the reconstruction are the imprints taken from the mesh, and the light red areas are the interpolation of the finger morphology. Right: schematic of the cycles of rainfall–evaporation events, indicating the interval time Δt between rain events.

evaporation whilst providing 3D visualisations of the flow patterns associated with deep drainage. Section 3 presents an analysis of the finger onset time, finger morphology and accumulation rate of water in the porous medium for different climate conditions ranging from humid to semi-arid. In Section 4, we discuss the implications of finger formation on evapotranspiration and the long-term effects on water availability in the context of potential global climate shifts.

2. Materials and methods

2.1. Experimental setup

The experimental setup consists of an acrylic cylindrical tube (diameter of 15 cm) that is packed with dry glass beads (particle size, d_p = 0.4–0.6 mm and an approximately Gaussian size distribution (Holtzman et al., 2012)) to a height of 30 cm, as illustrated in Fig. 1. The cell was vibrated by tapping on it around the sides to ensure a uniform packing. The beads are boro-silicate glass which is moderately wetting, with a contact angle on a flat surface of 50–60° as detailed in the Appendix. Air can escape from the bottom of the cell via a thin slit between the main cell and the bottom plate.

A rainmaker reproduces seasonal rainfall patterns. In our design, the rainmaker is a closed acrylic box (15 × 15 × 5 cm) with 102 needles protruding from the base (Fig. 1, inset). Although the rainmaker is a rectangular configuration the needles are fixed in a circular pattern to ensure an even distribution across the surface of the glass beads. The needles have an internal diameter of 2 mm and are placed with 1 cm spacing to replicate the size and spatial distribution of rain drops (Villermaux and Bossa, 2009). A plastic tube was attached to the top of the rainmaker and was connected to a large syringe. To simulate rainfall, water was drawn up into the box by the syringe through the needles. The rainmaker is then placed over the experimental cell. Precise volumes of water were manually injected over 1 min, after which, and during the evaporative intervals, the rainmaker is removed from the experimental cell. The short rain events are alternated with periods of intense 'sunlight' provided by a heat lamp (250 W). The volume of rainfall and the interval between rain events were varied.

The whole cell was placed on a platform which could pivot from its primary location, underneath the heat lamp, onto a balance. The mass of the cell was recorded before the experiment began, after each evaporation cycle, and after each rain event. Our experimental setup was designed to avoid disrupting the water–air interfaces (the mechanism moves in one smooth motion onto and off the balance), but measurements during the evaporative period were not taken as they would have interrupted the evaporative process and the movement of the cell could have unduly influenced the transport phenomena. Furthermore, it was the objective of this study to understand and replicate the overall process long term dynamics via many repeating rain cycles as other studies have already focused on the evaporation mechanism itself (Lehmann et al., 2008). It should also be noted that experiments were also conducted without forced evaporation in controlled laboratory conditions as a comparison to the forced evaporation case.

In order to provide insights into the 3D morphology of the wetting front, three disks of nylon mesh with nominal openings of 0.84 mm were inserted into the bead pack during the packing process at 5, 15 and 25 cm from the top of the cell. The tap water used to mimic rain was dyed with Allure red at a concentration of 2.4 g/l—this high dye concentration was selected so that the fluid created an unambiguous visible stain on the nylon mesh as it passed through. At the end of the experiment the mesh disks were removed and used to reconstruct the 3D morphology of the wetting front. By analyzing the fluid stains on the mesh cutouts at different depths it was possible to reconstruct the final finger morphology based on a reasonable interpolation for the regions between the mesh. In Fig. 2 one such reconstruction is shown; the dark areas represent the actual staining of the mesh by the red-dyed fluid and the lighter colors show a probable structure of the finger. The bottom of the cell is transparent and by visual inspection it was possible to determine when the front had reached the bottom. The end of the experiment is at least 5 rain cycles after the finger reached the bottom of the cell.

In the field, the sun is the driving force controlling PET. In the laboratory, we substitute this for a heat lamp. The lamp selected has a constant output of 250 W, and we conducted a calibration to determine that this provides a constant evaporation rate from a free water surface of 38 mm/day at a distance of 30 cm. This heat source and distance was used for all the experiments, so the PET is fixed in our experimental system.

2.2. Climate and soil conditions

Different climate conditions can be classified by the aridity index (Middleton and Thomas, 1997), which is defined as the ratio of mean annual precipitation (MAP) to potential evapotranspiration (PET) that aggregates maximum evaporation and transpiration if sufficient water were available,

Table 1
Aridity index of each climate class.

Aridity index, β	Climate class
< 0.03	Hyper Arid
0.03 – 0.2	Arid
0.2 – 0.5	Semi-arid
0.5 – 0.65	Dry sub-humid
> 0.65	Humid

Table 2
The precipitation magnitude of each rain event, P [mm], for each inter-event interval time Δt [min] and aridity index β.

		β			
		0.3	0.6	1.0	2.0
Δt [min]	180	2.10	4.21	–	–
	90	1.05	2.10	3.15	6.30
	30	0.35	0.70	1.05	2.10
	10	–	–	0.35	0.70

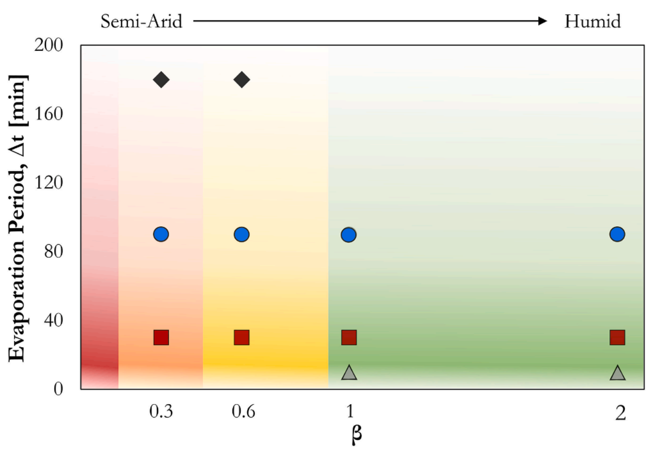

Fig. 3. Parameter space in our study, showing the experimental values of climatic conditions (aridity index β) and rainfall episodicity (inter-event period Δt).

$$\beta = \frac{\mathrm{MAP}}{\mathrm{PET}}. \tag{1}$$

Table 1 outlines the thresholds of β in the classification of regions of different aridity. In this study, we reproduce conditions corresponding to semi-arid ($\beta = 0.3$), dry sub-humid ($\beta = 0.6$) and humid environments ($\beta = 1, 2$).

Early works on the dynamics of gravity fingering have identified key parameters controlling the fingering process (Hill and Parlange, 1972; Parlange and Hill, 1976; Glass et al., 1989). In particular, they found that gravity fingering was strongly dependent on the flux ratio R_s [–], defined as the ratio of the infiltration rate R_f [m/s] to the saturated hydraulic conductivity,

$$R_s = \frac{R_f}{K_{\mathrm{sat}}}, \tag{2}$$

where $K_{\mathrm{sat}} = \rho g k/\mu$ [m/s], and where ρ [kg/m^3] is the fluid density, g [m/s^2] is the gravitational acceleration, μ [Pa.s] is the fluid viscosity and k [m^2] is the intrinsic permeability of the porous medium, which we estimate using the Kozeny–Carman (Kozeny, 1927; Carman, 1937) model, $k = (d_p^2/180)\phi^3/(1-\phi)^2$, where d_p is the mean grain diameter, and ϕ is the porosity. The value of the permeability estimated by the Kozeny–Carman equation is only an approximation, but one that has proven to be sufficient for the type of 3D glass-bead packs that we use here (Dalbe and Juanes, 2018). In the current experimental system, R_f plays the role of MAP in the field.

Rain events are inherently episodic, and the duration of a rain event can vary considerably depending on the time of year and geography of a given location, especially in semi-arid and arid regions (Snyder and Tartowski, 2006). In this study, we model rainfall episodicity by varying the volume of a rain event, V [m^3] and the time interval between rain events, Δt [s] (Fig. 2).

We define the MAP as

$$\mathrm{MAP} = \frac{P}{\Delta t}, \tag{3}$$

where $P = V/A$ is the intensity of the precipitation event, and A [m^2] is the cross sectional area of the cell.

To select appropriate values of Δt the experimental time must be scaled to the field. The characteristic time for water to flow through a given depth of porous medium, H, assuming fully-saturated conditions, is

$$t_c = \frac{H\phi}{K_{\mathrm{sat}}}. \tag{4}$$

Typical values of interest in a field setting (Wang et al., 2007) are $K_{\mathrm{sat}} = 5$ m/day, $H = 10$ m and $\phi = 0.45$, for which $t_{c,\mathrm{field}} \approx 0.9$ days. The values in our laboratory experiments are $K_{\mathrm{sat}} = 200$ m/day (using bead size of 0.5 mm), $H = 0.3$ m and $\phi = 0.4$, $t_{c,\mathrm{exp}} \approx 1.2$ min. Therefore, $\approx$1 day of field time corresponds to $\approx$1 min of experimental time.

We choose a fixed duration of rain events, equal to 1 min, and vary the time intervals between rain events, $\Delta t = 10$, 30, 90 and 180 min. Given Δt, one can determine the value of the rainfall precipitation per event P to achieve a given value of the aridity index β (Table 2). The range of variables explored in our study represents what was feasible in terms of injection volume V and reasonable experiment time constraints.

The parameter space for our experiment is shown in Fig. 3. The different climate conditions (arid, semi-arid, sub dry humid and humid) are shown, corresponding to vertical lines of constant values of β used in our study. In our study, the heat lamp has a fixed output and the distance to the top of the cell was also held fixed across experiments, so PET is constant (38 mm/d). We vary the aridity conditions by varying the average flux ratio R_s, and the rainfall episodicity by varying the inter-event time Δt. The 12 conditions investigated experimentally (see Table 2) are indicated with a symbol.

3. Results

In this section, we report the results of our laboratory experiments on gravity fingering in the presence of surface evaporation and without forced evaporation as a comparative case, including: (1) finger size and morphology; (2) onset time of the fingering instability; (3) amount of evaporation that takes place; and (4) partition between evaporation and deep drainage.

3.1. Finger size and morphology

The finger characteristics have been analyzed exhaustively in studies *without* evaporation, in quasi-2D experiments (Diment and Watson, 1985; Glass et al., 1989; Selker et al., 1992; Selker et al., 1992; Lu et al., 1994; Bauters et al., 1998; Bauters et al., 2000; Yao and Hendrickx, 2001; Sililo and Tellam, 2000; Flekkøy et al., 2002; Wang et al., 2004; Wei et al., 2014) and some 3D experiments (Yao and Hendrickx, 1996; Glass et al., 1990). Here, we study the finger size in the presence of evaporation.

In all of the 12 cases, with or without evaporation, one large dominant gravity finger formed. Once the front becomes unstable, the fastest-growing finger creates a preferential channel through which most (if not all) of the fluid drains, inhibiting the growth of subsequent fingers in a

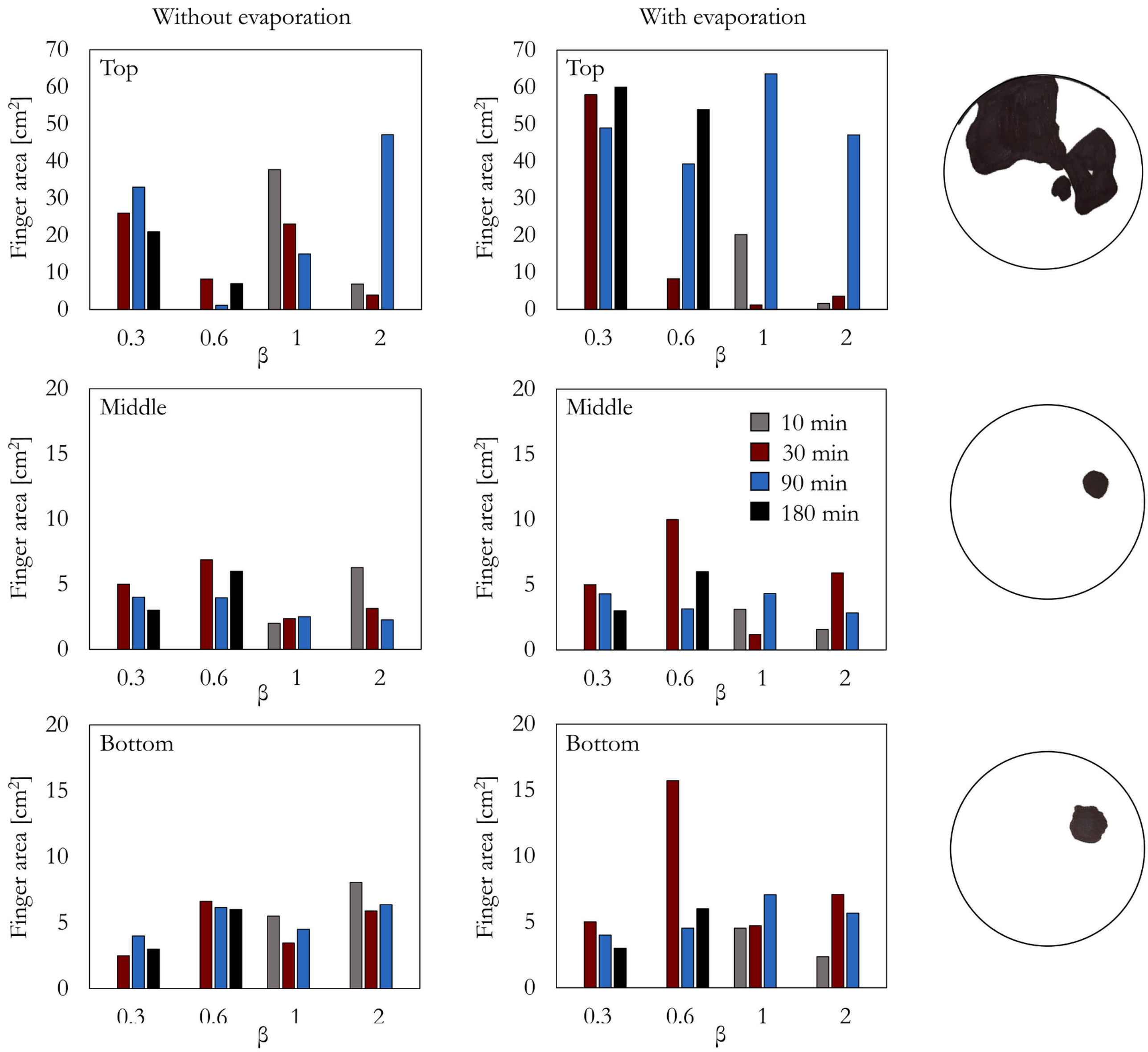

Fig. 4. Finger size at the top (5 cm), middle (15 cm) and bottom (25 cm) mesh locations. The results are grouped by β; for each value of β, different colors represent different Δt. To the right is an example of the mesh image after binarization for the case with evaporation at $\beta = 1$ and $\Delta t = 90$ min, as indicated with an asterisk on the bar chart.

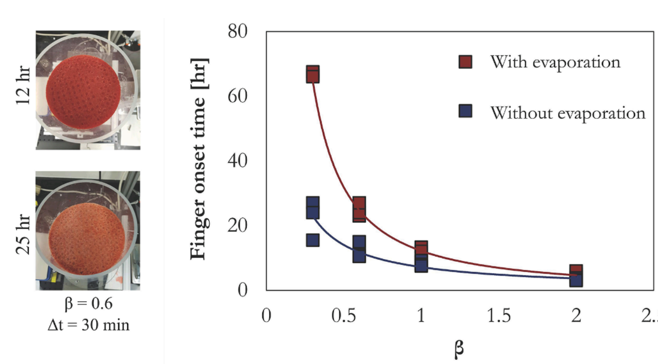

Fig. 5. Two birds-eye photographs of the cell before (top) and after (bottom) a finger has formed for an experiment with evaporation. The graph shows the finger onset time as a function of β for the case with evaporation (red symbols) and without evaporation (blue symbols).

winner-takes-all process.

In Fig. 4 we show an example of the finger structure obtained from the stains in the nylon mesh cutouts at three different depths from the surface of the bead pack: top (5 cm), middle (15 cm) and bottom (25 cm). From the binarized image of the cross sections, we determine the area of the finger at each depth. In general, the largest finger cross-section is observed at the top and for the case with evaporation. The wetting front has penetrated far enough into the cell and has come in contact with the mesh at several points, resulting in a large finger size. In the case with evaporation, the finger size is larger for small β, which indicates that the fluid front extends further into the cell because a greater area of the wetting front contacts the mesh.

Although it was not possible to precisely quantify the depth of the wetting front within the cell we did observe from the outside of the cell that in cases without evaporation the front extended approximately 2.5 cm and with evaporation the front extended approximately 5 cm. Evaporation itself occurs at a sharp surface but over time the location of this front will propagate downwards into the sand pack if, as in this case, the evaporation rate outpaces the rate of infiltration (Gran et al., 2011).

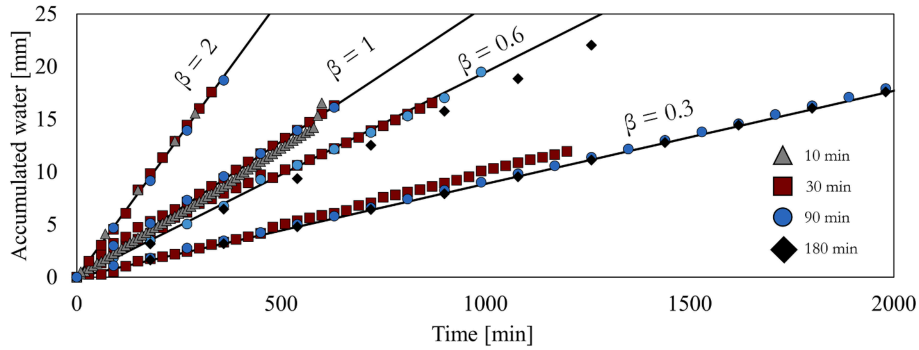

Fig. 6. Cumulative water in the cell as a function of time for different aridities β (= 2, 1, 0.6 and 0.3), and for different interval times Δt (= 10 min (triangles), 30 min (squares), 90 min (circles) and 180 min (diamonds)) for the case without evaporation. For each value of β, the theoretical accumulation assuming a closed system is plotted with a straight line.

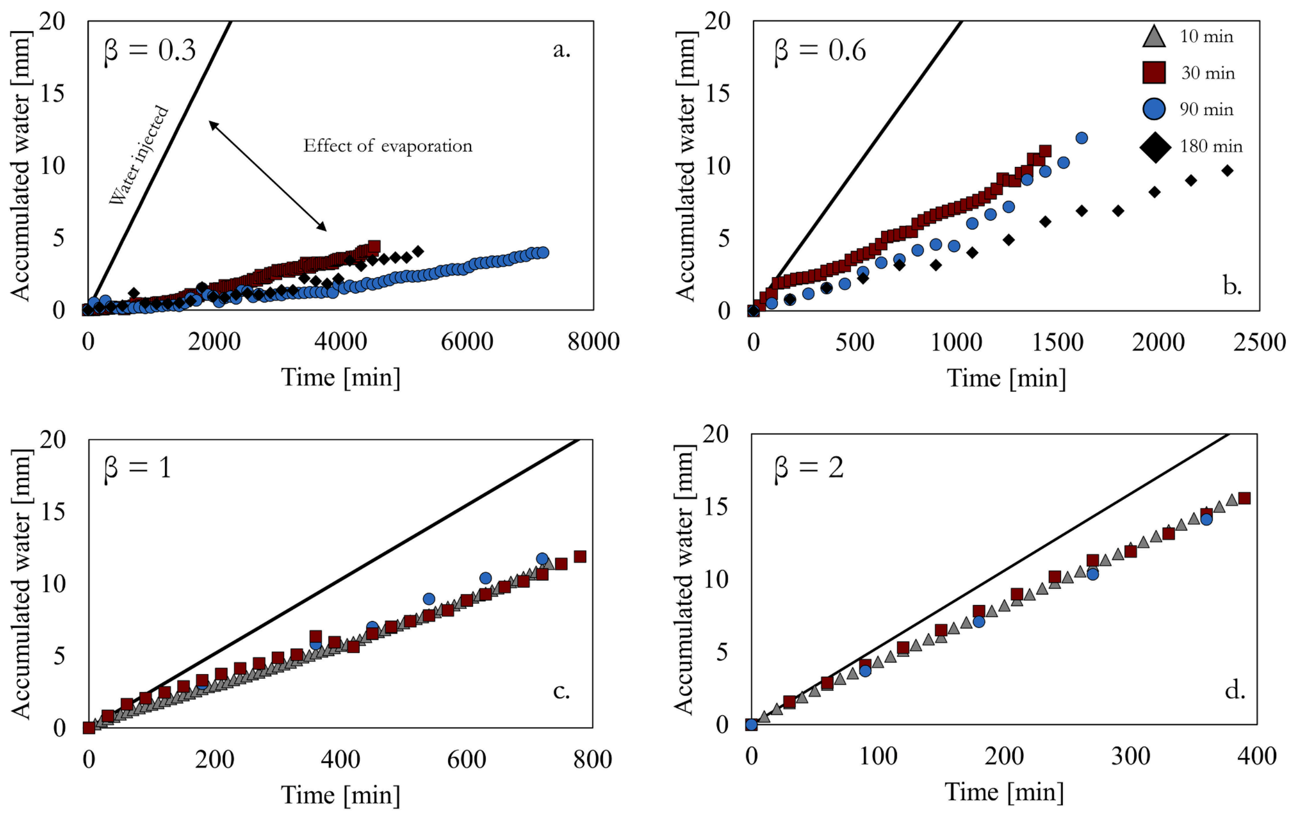

Fig. 7. Cumulative water in the cell as a function of time for: (a) $\beta = 0.3$; (b) $\beta = 0.6$; (c) $\beta = 1$; and (d) $\beta = 2$, for different interval times Δt (= 10 min (triangles), 30 min (squares), 90 min (circles) and 180 min (diamonds)) for the case with evaporation. For each value of β, the theoretical accumulation assuming a closed system is plotted with a straight line.

The depth to which this front extends downwards is often referred to as the 'evaporative zone' and our observation supports the hypothesis that the characteristic length of this zone is determined by soil properties (Lehmann et al., 2008; Shokri et al., 2010). Within this zone less water accumulates near the surface but over time a sufficient amount of water can propagate beyond this zone in order for a saturation overshoot at the wetting front to occur that is responsible for triggering the fingering instability (DiCarlo, 2004). Therefore, the front propagates deeper into the cell, over a longer period of time, thus forming the larger finger structure at the top mesh location.

Between the top and the middle slices the front has channelized into one dominant pathway. At the middle location, the finger area is consistently ≈ 5 cm^2, as this is controlled primarily by the fluid and porous-medium properties, both of which are constant in our experiments. However, if this study were extended to different particle sizes, then a trend in finger morphology would be expected accordingly. We often observe that the finger is at its smallest size in the middle of the cell, and that its area increases as it reaches the bottom, suggesting a boundary effect (Fig. 4).

A challenging factor to account for is the potential impact of temperature on finger morphology. Higher temperature (which would be expected in the experiments with evaporation) would lead to a decrease in both viscosity and surface tension. A lower viscosity implies a larger saturated conductivity and, in turn, a lower flux ratio R_s. A lower surface tension implies a smaller capillary height. Both of these trends would result in thinner gravity fingers (Glass et al., 1989; Wei et al., 2014; Beljadid et al., 2020).

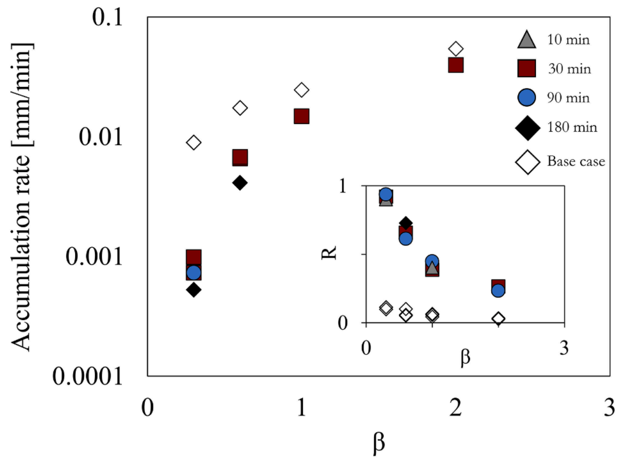

Fig. 8. Rate of accumulation of water, r_{accum}, for each case without surface evaporation (empty symbols) and with evaporation (filled symbol) as a function of β, indicating the rainfall interval time Δt of 30 min (red squares), 90 min (blue squares) and 180 min (black diamonds) for $\beta < 1$. Inset: Evaporative fraction R for experiments without evaporation (empty symbols) and with evaporation (filled symbols) as a function of β.

3.2. Onset time of gravity fingering

After many injection cycles, a gravity finger ultimately formed in each of the experiments. The onset time is the time of the rain event that triggered the finger. It was possible to infer the formation of a gravity finger by a color change at the surface as the fluid drained towards the bottom. The top surface of the cell becomes visibly dry and pale once as the finger is formed (Fig. 5). In addition to the color change in the view from the top, and to confirm that a finger had formed, the bottom of the cell is transparent so the finger could be visualized quickly reaching the bottom of the cell.

In Fig. 5 we plot the finger onset time as a function of β, for the base case without forced evaporation (blue symbols) and with forced evaporation (red symbols). In both cases, the onset time increases with decreasing β, and the trends follow a nonlinear, inverse power law relationship. Under humid conditions ($\beta = 2$), the onset time is similar for all cases ($\approx$ 2 h). However, the trends between the experiments with and without evaporation diverge as β decreases and the onset time for the case with evaporation is increasingly longer compared to the case without evaporation. Under semi-arid conditions ($\beta = 0.3$), the onset time is 3 times greater for the case with evaporation. Although further experiments are required, this result suggests that at sufficiently arid conditions ($\beta \rightarrow 0$), gravity fingers will not form. Indeed, at extremely high aridities, one expects that all of the rainfall water will evaporate before it can pile up to sufficiently high saturation that it can trigger the fingering instability-a condition that is known to be necessary for gravity fingering (DiCarlo, 2004; DiCarlo, 2007).

3.3. Evaporation at different aridity indices

The mass of the cell was measured at the start of each experiment, and then after each rain event and again after every evaporation period, where the period of evaporation varied from 10 to 180 min.

In Fig. 6 we plot the accumulated water in the cell scaled by the cross sectional area of the cell as a function of time for the system without forced evaporation; however, this is not a closed system so there was some minimal evaporation to the atmosphere in the ambient laboratory conditions. The solid lines indicate the theoretical accumulation assuming a closed system. As β decreases, the water accumulation is slower. The close agreement between the experimental points and the theoretical line for each value of β indicates that, indeed, the water losses are minimal in the experiments without evaporation and serve as control experiments. It should be noted that in the case without forced evaporation the label with the β value is merely an indication to facilitate the comparison with the corresponding experiments with forced evaporation.

In Fig. 7 we present the experiments with evaporation, grouped by the value of the aridity index β. The difference between experimental points and the theoretical line indicates the amount of evaporation that has taken place. In the experiments with evaporation, for the semi-arid conditions ($\beta = 0.3$ and $\beta = 0.6$), the rate of water accumulation in the system exhibits a dependence on rain inter-event time Δt. As the interval between rain events lengthens, the rate of accumulation decreases, indicating increased evaporation. In contrast, for humid conditions ($\beta = 1$ or $\beta = 2$), the accumulation rate is independent of Δt.

To compare these trends more clearly, we fit a straight line to the data in Figs. 6 and 7 before the finger forms, and the slope represents the rate of accumulation of water for each case. Fig. 8 shows the accumulation rate for the case without evaporation (empty symbols), $r_{without}$, and for cases with evaporation (filled symbols), r_{with}, as a function of β. For $\beta > 1$ the rate of change is very similar for different interval times, so a single point is plotted for clarity.

With evaporation, in general, as β decreases, the conditions become dryer and water accumulates in the system at an increasingly slower rate. The rate of accumulation for $\beta = 0.3$ is two orders of magnitude lower than for $\beta = 2$. We quantify this marked disparity in the system's response by means of the evaporative fraction R, defined as the ratio between the rates of mass evaporated and mass infiltrated at the surface, before the finger forms:

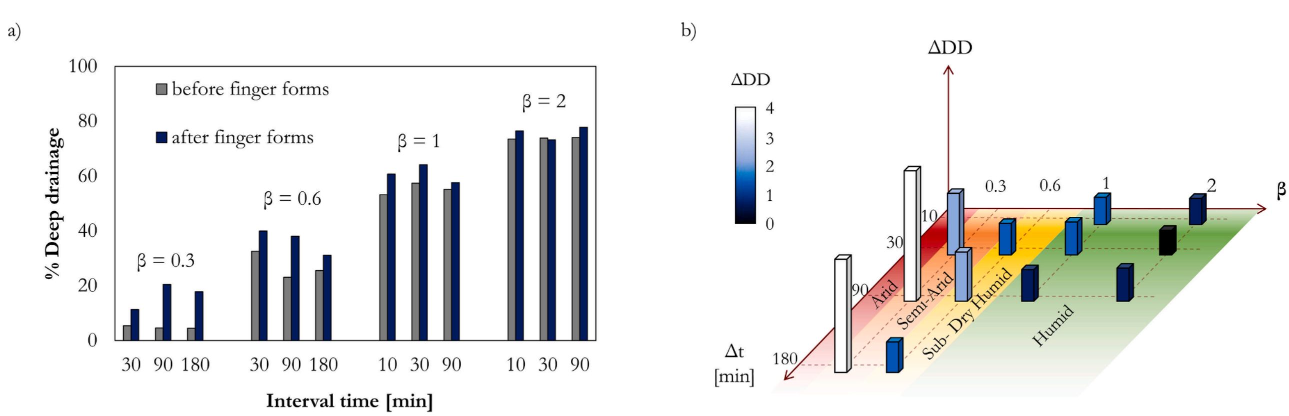

Fig. 9. Left: Average deep drainage as a percentage of injection volume before and after a gravity finger forms. The results are shown for each interval time Δt, grouped by β. Right: Change in deep-drainage fraction, ΔDD, indicating the impact of gravity fingering on deep drainage for different climate conditions (β and Δt) investigated experimentally.

$$R := \frac{r_{\text{evap}}}{r_{\text{infil}}} = 1 - \frac{r_{\text{accum}}}{r_{\text{infil}}}. \quad (5)$$

The evaporative fraction for different aridity indices β is shown in Fig. 8inset).

3.4. Impact of gravity fingering on evaporation and deep drainage

We perform a mass balance throughout the experiment to separate the contribution of infiltrated water to evaporation and to deep drainage, *before* and *after* the finger forms. The time-average fraction of infiltrated water that does *not* evaporate is shown in Fig. 9left), comparing these fractions before and after the onset of fingering, for the different values of β and Δt in the experiments.

To gain further insight into the impact of gravity fingering on coupled infiltration–evaporation, we define the deep-drainage fraction,

$$R_{\text{DD}} = 1 - R = \frac{r_{\text{accum}}}{r_{\text{infil}}}. \quad (6)$$

As expected, R_{DD} increases with increasing β (that is, as the conditions become more humid) because the evaporative fraction is decreasing (Fig. 8). For humid conditions ($\beta = 1$ and 2), the values of R_{DD} before and after the finger forms are very similar—indicating that fingering has, in these cases, a small contribution to the infiltration–evaporation dynamics. For sufficiently dry conditions ($\beta = 0.6$ and especially 0.3), however, we observe a significant increase in R_{DD} from before to after fingering; more infiltration water bypasses evaporation and contributes instead to deep drainage.

To quantify this change in R_{DD} as the system transitions from compact infiltration to fingered infiltration, we define

$$\Delta\text{DD} = \frac{R_{\text{DD,finger}} - R_{\text{DD,compact}}}{R_{\text{DD,compact}}}. \quad (7)$$

This metric allows us to synthesize the impact of gravity fingering on deep drainage for different climate conditions (Fig. 9right)), illustrating that even for moderate aridity (semi-arid conditions), gravity fingering results in a fourfold increase in deep drainage.

4. Discussion and conclusions

We have developed a novel experimental strategy to investigate the role of gravity fingering during water infiltration in a 3D porous medium. While previous studies had already indicated the prevalence of gravity fingering in the laboratory and in the field, here we build on this work to include cycles of rain and periods of evaporation. We find that—especially in dry conditions—gravity fingers exert a powerful control on evapotranspiration.

Our experimental results show that gravity fingering enables infiltration water to bypass the evaporative zone and penetrate deep into the subsurface. Given that the experimental conditions were designed to reproduce climate conditions in the field, these results may have significant implications for the partition between evaporation and deep drainage in water-stressed ecosystems. In particular, we have studied the impact of the aridity index ($\beta = \text{MAP/PET}$) and the rain episodicity (as measured by the inter-event time Δt) on the coupled infiltration–evaporation dynamics. We find that, for moderately arid climates, deep drainage can increase multiple-fold in the presence of gravity fingering. We also find that this is exacerbated when rain events are more intense and less frequent (Fig. 9); more intense rainfall events between long periods without rainfall (e.g., $\beta = 0.3, \Delta t = 90$ min) results in less evaporation losses and more deep drainage compared to several small rainfall events with short waiting periods between rainfall events (e.g., $\beta = 0.3, \Delta t = 30$ min). These results are consistent with recent field work (Lehmann et al., 2019) who, based on lysimeter observations and a surface evaporation capacitor (SEC) model, find that deep drainage (i.e., leakage from the capacitor) is enhanced for regions with higher rainfall variability and propensity of large rainfall events.

It has been previously reported that an increase in precipitation variability will shift the subsurface water-availability locus from the near-surface to greater depth (Sala et al., 2015). This phenomenon has been attributed to the disproportionate increase in the water accumulation at depth following large volumes of rainfall, where ideas such as 'soil memory' have been proposed to explain how soil–water saturation profiles persist long after the rain event (Sala et al., 2015). Our experimental results provide a mechanistic explanation for this effect, and show that the longer between rain events and the larger the volume of rainfall, the greater the effect on deep drainage (Fig. 9), albeit after a longer onset time (Fig. 5). 'Soil memory' is furnished by flow channeling, where future rainfall events have a higher probability of feeding wet soil via established gravity fingers, thus allowing for faster deep percolation.

Although our study makes some important simplifications (for example, we do not account for surface runoff and vegetation), it highlights that the emergence of subsurface channelization of rainwater can impact water resources to a degree, and on a timescale, of ecosystem relevance. Our results therefore suggest that the subtle interplay between gravity fingering and evapotranspiration may be essential to explain the structure and resilience of water-stressed ecosystems, influencing their response to climate variability.

Plain language summary

The movement of water through soil is a fundamental process of the hydrologic cycle, determining the partition of infiltrated water into evaporation, uptake by plant roots, and deep drainage into groundwater reserves. Here we use a novel experimental method to study coupled infiltration–evaporation in a 3D porous medium, reproducing field conditions of aridity and rainfall episodicity. We show that evaporation is hindered, and deep drainage significantly enhanced, by the formation of gravity fingers—preferential columnar paths that allow water to percolate quickly in subsequent rainfall events, thus bypassing the evaporative zone in the soil. These findings explain previously hypothesized 'soil memory' effects in arid regions, and suggest that fingered infiltration may be a critical determinant of the function and resilience of water-stressed ecosystems.

CRediT authorship contribution statement

Rebecca Liyanage: Methodology, Formal analysis, Investigation, Data curation, Writing - original draft. **Ruben Juanes:** Conceptualization, Methodology, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

We characterize the wettability of the beads in our system by placing a small drop of DI water on a glass surface (McMaster, Borosilicate Glass Sheet) in ambient air. The droplet volume is between 0.5 and 1 microliter. We image the water drop with a goniometer (model 250, Rame-hart), which measures the water/air contact angle via the DROPimage software. This process is repeated with five different water droplets as shown in Fig. 10. The average contact angle is $54/+-6^o$.

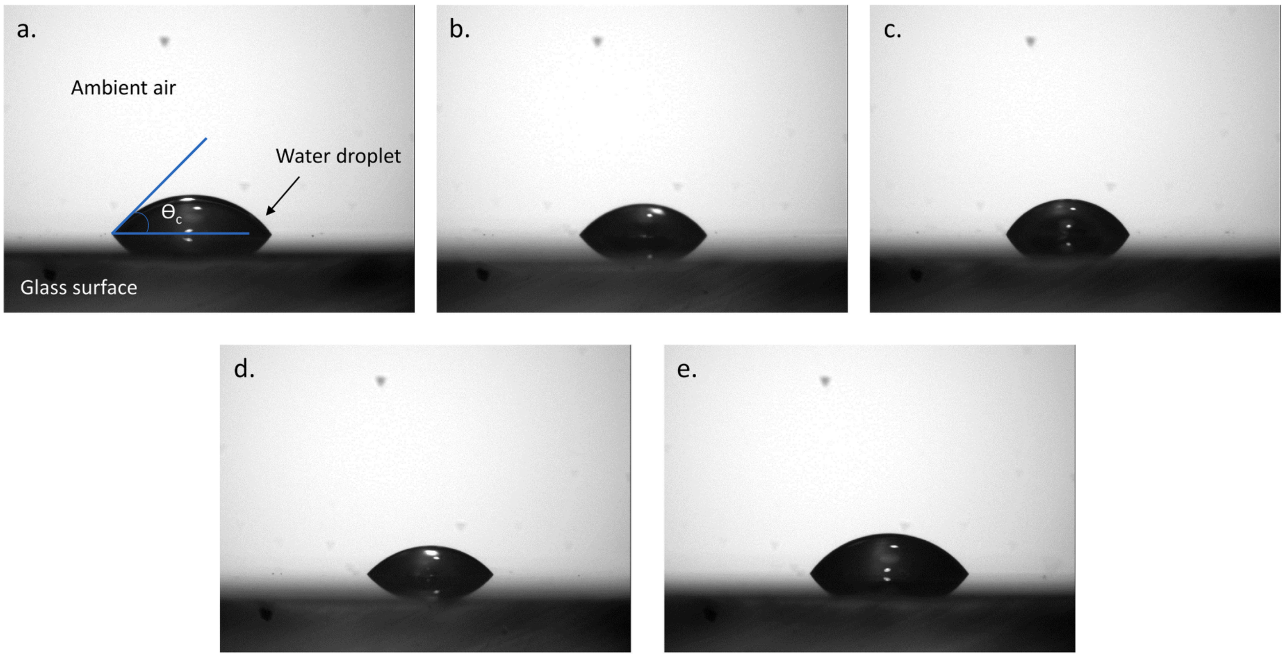

Fig. 10. Five individual water droplets (a-e) on a borosilicate glass surface in ambient air. The contact angle measured with a goniometer (model 250, Rame-hart) via the DROPimage software.

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